Tutorial 3

Ex. 8.5 Q45

$$\int \frac{1}{x^{3/2} - \sqrt{x}}\, dx$$

$u = \sqrt{x} \Rightarrow x = u^2$

$du = \frac{1}{2\sqrt{x}}\, dx$

$$= \int \frac{1}{\boxed{\sqrt{x}}(x-1)}\, dx$$

$$= 2\int \frac{1}{u^2 - 1}\, du$$

$$= 2\int \frac{1}{(u-1)(u+1)}\, du$$

$$\frac{1}{(u-1)(u+1)} = \frac{A}{u-1} + \frac{B}{u+1}$$

Q46 $\displaystyle\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$

~~Hint: $x = u^6$~~ $\rightarrow$ $x^{m/n} \rightarrow$ Try $u = x^{1/n}$

$u = \sqrt{x} \Rightarrow x = u^2 \Rightarrow x^{1/3} = u^{2/3}$

$du = \frac{1}{2\sqrt{x}}\,dx$

$$= 2\int \frac{1}{u^{2/3}-1}\,du$$

$t = u^{1/3} \Rightarrow u^{2/3} = t^2$

$dt = \frac{1}{3}u^{-2/3}\,du$

$= \frac{1}{3}\left(\boxed{u^{1/3}}\right)^{-2}\,du$

$= \frac{1}{3}t^{-2}\,du$

$du = 3t^2\,dt$

$$= 2\int \frac{3t^2}{t^2-1}\,dt$$

$$= 6\int \frac{t^2}{t^2-1}\,dt$$

$$\frac{t^2}{t^2-1} = \frac{\boxed{t^2-1}+1}{t^2-1} = 1 + \frac{1}{t^2-1}$$

$$\frac{1}{t^2-1} = \frac{1}{(t-1)(t+1)} = \frac{A}{t-1} + \frac{B}{t+1}$$

Q47 $\int \frac{\sqrt{x+1}}{x}\,dx$

① $u=\sqrt{x}$ $u=x^{1/n}$

② $u=\sqrt{\cdots}$

$u=\sqrt{x+1}$

$u^2=x+1$

$x=u^2-1$

$dx=2u\,du$

$$=\int \frac{u\cdot 2u\,du}{u^2-1}$$

Rational function

Try Q48, Q53, Q54

Q50

$$\int \frac{1}{x^6(x^5+4)}\,dx \cdot \frac{x^4}{x^4}$$

$$= \int \frac{x^4}{x^{10}(x^5+4)}\,dx$$

$$= \int \frac{1}{5}\cdot\underbrace{\frac{1}{u^2(u+4)}}_{\text{P.F.}}\,dx$$

$u = x^5 \Rightarrow du = 5x^4\,dx$

$x^{10} = u^2$

Q49

$$\int \frac{1}{x(x^4+1)}\,dx \quad \text{Hint: Mult. by } \frac{x^3}{x^3}$$

$$\| \quad \frac{x^3}{x^4(x^4+1)} \rightarrow d(x^4) = 4x^3\,dx$$

$$\int \frac{1}{x^m(x^n+4)}\,dx \qquad x^{m+n-1} = x^{\ell n} = (x^n)^\ell$$

a multiple of n

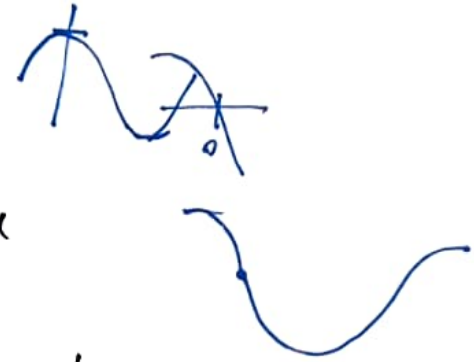
If $m+n-1$ is a multiple of n, then try mult. the integral by $\frac{x^{n-1}}{x^{n-1}}$.

Ex. 8.8

① Just evaluate

② Direct Comparison Test $\rightarrow$ $\int_a^{\boxed{\infty}} f(x)\,dx$

③ Limit Comparison Test $\nearrow$



Q40 $\displaystyle\int_0^{\pi/2} \cot\theta \, d\theta$

$\cot\theta = \dfrac{1}{\boxed{\tan\theta}}$

$\tan\theta = 0$ when $\theta = 0$

$$= \lim_{c\to 0^+} \int_c^{\pi/2} \cot\theta \, d\theta$$

$$= \lim_{c\to 0^+} \Big[\ln|\sin\theta|\Big]_c^{\pi/2}$$

$$= \lim_{c\to 0^+} \Big(\underbrace{\ln\left|\sin\tfrac{\pi}{2}\right|}_{0} - \ln|\sin c|\Big)$$

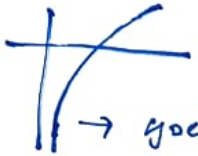
$$= \infty$$

$$\int \cot\theta \, d\theta = -\ln|\csc\theta| + C$$

$$= \ln|\sin\theta| + C$$

$$= \ln\left(|\csc\theta|^{-1}\right)$$

$\ln x$



$\rightarrow$ goes to $-\infty$ when $x \to 0^+$

$\therefore$ The improper integral $\int_0^{\pi/2} \cot\theta \, d\theta$ diverges.

<u>Q48</u> $\displaystyle\int_0^2 \frac{dx}{\underbrace{1-x}_{\boxed{x=1}}} = \underline{\int_0^1 \frac{dx}{1-x}} + \int_1^2 \frac{dx}{1-x}$

$b \to 1^-$

$1-b \to 0$

$\ln|1-b| \to -\infty$

$-\ln|1-b| \to +\infty$

$$\int_0^1 \frac{dx}{1-x} = \lim_{b\to 1^-} \int_0^b \frac{dx}{1-x}$$

$$= \lim_{b\to 1^-} \Big[-\ln|1-x|\Big]_0^b$$

$$= \lim_{b\to 1^-} \left(\cancel{-\ln|1-b|} + \ln|1-b| + \ln 1\right)$$

$$= \infty$$

Since $\int_0^1 \frac{dx}{1-x}$ diverges, $\int_0^2 \frac{dx}{1-x}$ diverges.

Q59 $\displaystyle\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$

$-1 \leq \cos x \leq 1$

$1 \leq 2+\cos x \leq 3$

$\boxed{\frac{1}{x}} \leq \frac{2+\cos x}{x} \leq \frac{3}{x}$

$\displaystyle\int_{\pi}^{\infty} \frac{1}{x}\,dx \leq \int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$

Since $\displaystyle\int_{\pi}^{\infty} \frac{1}{x}\,dx$ diverges,

$\displaystyle\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$ diverges

by Direct Comparison Test.

$$\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx = \underbrace{\int_{\pi}^{\infty} \frac{2}{x}\,dx}_{\text{diverges.}} + \int_{\pi}^{\infty} \frac{\cos x}{x}\,dx$$

Since $\displaystyle\int_{\pi}^{\infty} \frac{2}{x}\,dx$ diverges, $\displaystyle\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx$ diverges.

Wrong!

$$\int_{a}^{\infty} \frac{1}{x} - \frac{1}{x+1}\,dx = \int_{a}^{\infty} \frac{1}{x}\,dx - \int_{a}^{\infty} \frac{1}{x+1}\,dx$$

Q66 $\displaystyle\int_1^\infty \frac{1}{e^x - 2^x}\,dx$

$e > 2$

$\boxed{e^x} >> 2^x$

Let $f(x) = \dfrac{1}{e^x - 2^x}$, $g(x) = \dfrac{1}{e^x}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\frac{1}{e^x - 2^x}}{\frac{1}{e^x}} = \lim_{x\to\infty} \frac{e^x}{e^x - 2^x}$$

$$= \lim_{x\to\infty} \frac{1}{1 - \underbrace{\left(\frac{2}{e}\right)^x}_{<1}}$$

$$= 1 > 0$$

$\displaystyle\int_1^\infty \frac{1}{e^x}\,dx = \cdots$ (Converges)

Since $\int_1^\infty \frac{1}{e^x}\,dx$ converges, $\int_1^\infty \frac{1}{e^x - 2^x}\,dx$ converges by Limit Comparison Test.

Q53 $\dfrac{1}{\sqrt{v-1}} \sim \boxed{\dfrac{1}{\sqrt{v}}}$

Q55 $\dfrac{1}{\sqrt{x^6+1}}$

$\sim \dfrac{1}{\sqrt{x^6}} = \boxed{\dfrac{1}{x^3}}$

Q57 $\dfrac{\sqrt{x+1}}{x^2}$

$\sim \dfrac{\sqrt{x}}{x^2} = \boxed{\dfrac{1}{x^{3/2}}}$